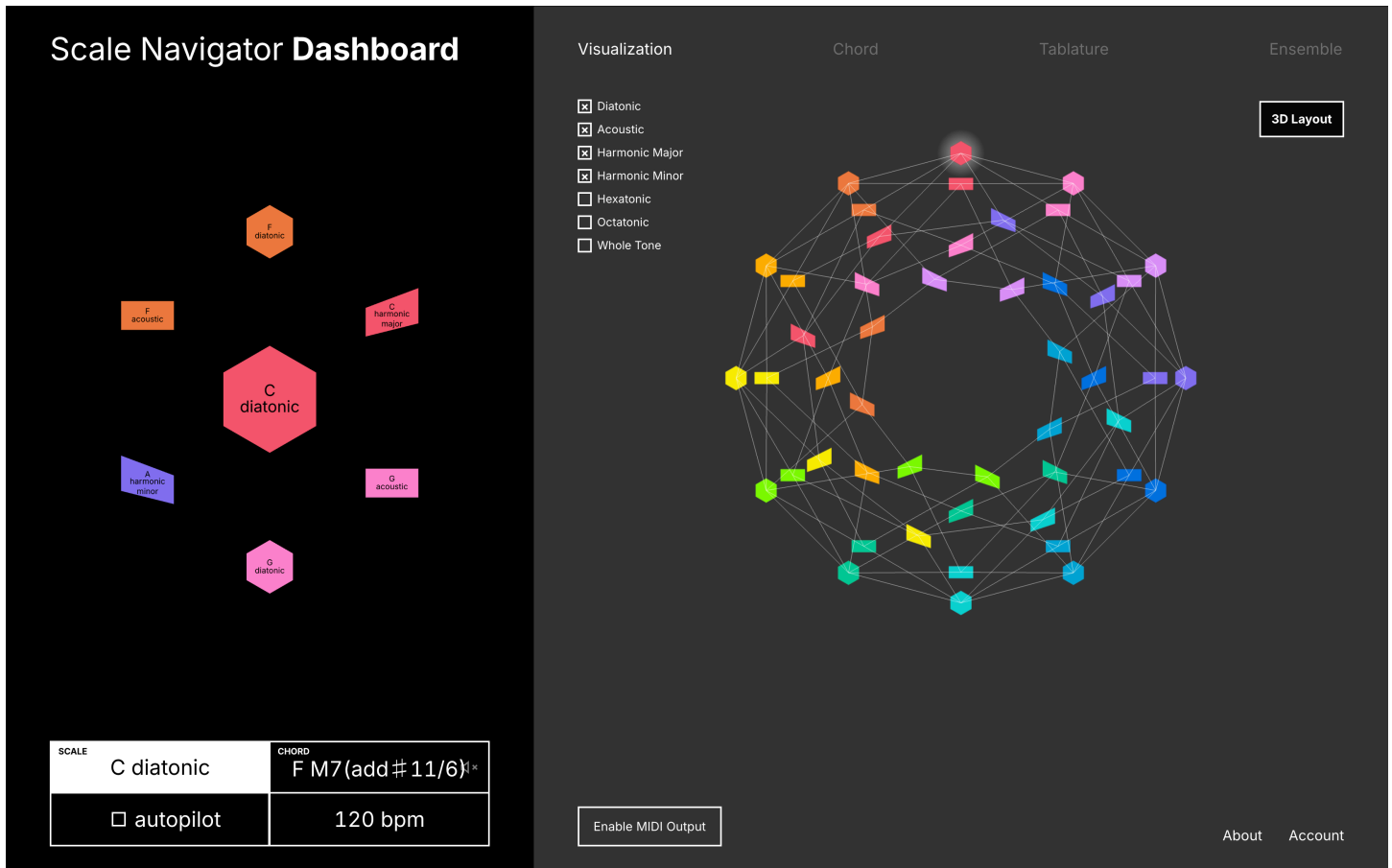


NATHAN TURCZAN

Composer / Product Designer / Creative Technologist

Los Angeles · nathanturczan@gmail.com · nathanturczan.com

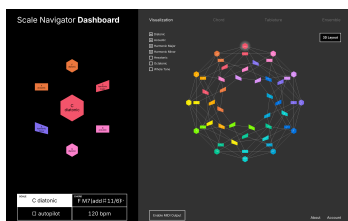


Building instruments that let people navigate harmony as interactive space.

I design and build software instruments that transform abstract music theory into navigable, playable environments. My work spans DAW plugins, web applications, mobile apps, networked performance systems, and interactive installations.

Currently developing the **Scale Navigator** ecosystem: harmonic middleware that coordinates your entire session from one interface. The visualization above maps 57 scales into a network where adjacent nodes differ by a single note, enabling smooth voice leading between any two harmonic states.

SELECTED PROJECTS

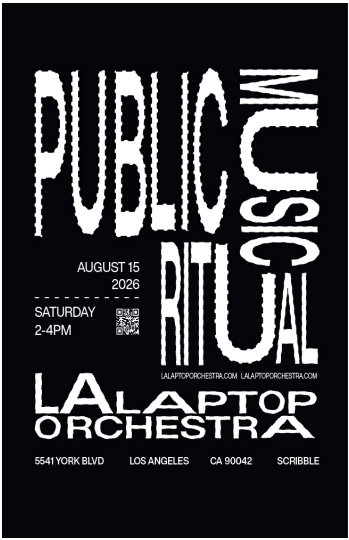


Scale Navigator Dashboard

2021–Present

PROBLEM: Producers lack a unified way to coordinate harmony across DAW, plugins, and instruments. Scale changes require manual updates to multiple tools.

BUILT: Cross-platform harmonic middleware (Web, iOS, Android, VST3/AU) that broadcasts scale and chord state via MIDI. One source of truth for your session's harmony. Free on all platforms.

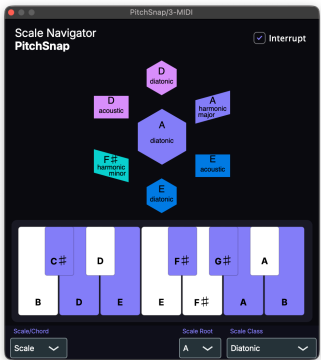


LA Laptop Orchestra

2026

PROBLEM: Experimental music communities lack accessible entry points. Laptop orchestras historically require technical expertise, excluding casual participants.

BUILT: Participatory public music ritual where strangers join a shared harmonic field via QR code. Phones, laptops, and acoustic instruments all connect to the same harmony: no wrong notes possible. Launching August 2026 at Scribble, Los Angeles.



Tonalign

2024–Present

PROBLEM: MIDI pitch correction tools don't exist. Producers can autotune vocals but not MIDI: they must manually correct notes outside the current scale.

BUILT: Polyphonic MIDI pitch correction plugin (VST3/AU). Routes incoming notes to nearest scale degree in real time. Try chord ideas in any scale: hear what works instantly.



SOULS (with Sia & David O'Reilly)

2022

PROBLEM: Interactive NFT characters needed ambient soundscapes that felt alive and responsive, not static audio loops.

BUILT: Procedural audio system generating unique ambient compositions for 10,000 collectible characters. Each SOUL has its own evolving sonic identity. Featured in Forbes, Designboom, OpenSea.

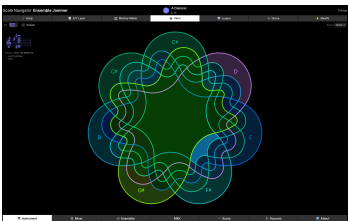


Modal Intersections

2018

PROBLEM: Audiences experience generative music passively. Interactive installations rarely give meaningful musical control to untrained participants.

BUILT: Audiovisual installation where audience movement controls real-time harmonic transitions. Player pianos respond to gesture; projections visualize shifting modal relationships. Exhibited at CalArts CEAIT Festival and REDCAT Gallery.



Ensemble Jammer

2024–Present

PROBLEM: Remote musicians can't improvise together without latency issues and harmonic chaos.

BUILT: Networked instrument app (Web, iOS) where host controls harmony, players respond with constrained instruments: Touch Harp, Drone, Sequencer. Firebase real-time sync. No wrong notes. Used for sound baths and remote jams.

ADDITIONAL WORK

SNaPS Plugin 2024 · Sampler/VST3

Harmony-aware granular sampling with 8 particle emitters. Co-developed with Nathan Ho.

Early Downhome Blues 2024 · Web

Interactive blues melody generator based on Jeff Todd Titon's musicology research.

Pandiatonic Autoharp 2021 · Instrument

Modified autoharp retuned for pandiatonic exploration: all seven diatonic notes.

To A Wild Rose... 2020 · Album

14 algorithmic remixes of Edward MacDowell via sample rate modulation.

Scale Awareness Bridge 2024 · M4L

Syncs Scale Navigator to Ableton's native scale mode and Push.

Voice Leader 2023 · M4L

Four-voice SATB harmony from single notes, based on Huron's Bach/Haydn analysis.

BACKGROUND

MFA Music Technology — CalArts, 2018

BA Vocal Performance — UC Santa Cruz, 2013

Thesis: Scale Navigator interface design based on Dmitri Tymoczko's voice-leading geometry research (NIME 2019 paper).

CTO and Co-founder, Chordas Collective (choral tech).

SKILLS

Languages: JavaScript/TypeScript, C++, Python, Max/MSP, SuperCollider

Frameworks: React, JUCE, Tone.js, Firebase, Capacitor

Tools: Ableton Live, Figma, Git, LaTeX, LilyPond

Domains: Audio plugin dev, real-time web audio, MIR, networked performance, interactive installation

Available for creative technology roles, interactive sound design, and music technology consulting.

nathanturczan@gmail.com · **nathanturczan.com** · **github.com/nathanturczan**